Quad HDMI test pattern generator — FPGA main board

USB-C 5 V -> bucks -> XC7A35T-1FTG256C (QSPI flash, JTAG, 100 MHz) --J1/J2 sockets--> quad_hdmi_tx carrier (4x ADV7513 -> 4x HDMI) stacked on top

Power: USB-C 5 V in, 1.0 V / 1.8 V / 3.3 V bucks

File: power.kicad_sch

Carrier board sockets J1/J2 (mate quad_hdmi_tx J7/J8)

File: carrier_if.kicad_sch

FPGA I/O banks 14/15/34/35

File: fpga_io.kicad_sch

User I/O: LEDs, DIP switch, reset button, PMOD

File: io_misc.kicad_sch

FPGA configuration, power pins, QSPI flash, JTAG, clock

File: fpga_cfg.kicad_sch

Quad HDMI test pattern generator — FPGA main board (XC7A35T, carrier stacks on top)

gyuhooo

Sheet: /

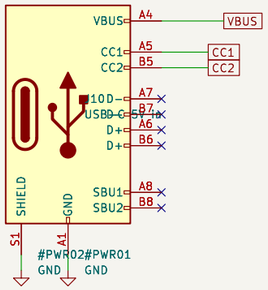
File: fpga_main.kicad_sch

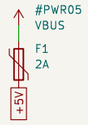
Title: FPGA main board (XC7A35T-FTG256) — root

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USB-C 5 V (2 A) -> PTC fuse -> +5V. Three synchronous bucks (AP63200 adj. / AP63203 3.3 V fixed, 2 A): +1V0 (VCCINT/VCCBRAM), +1V8 (VCCAUX), +3V3 (VCCO, I/O).

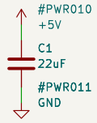
Power-on order per UG483 (VCCINT -> VCCAUX -> VCCO): U3 EN direct, U4 EN delayed ~30 ms (100k + 1uF), U5 EN delayed ~60 ms (100k + 2.2uF).

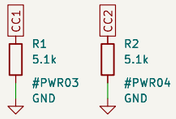


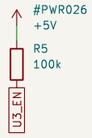


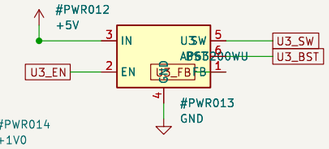


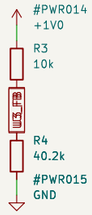


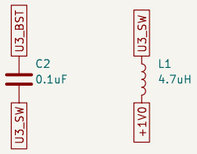




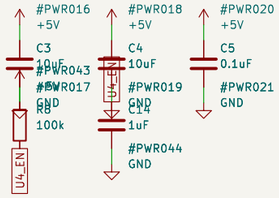


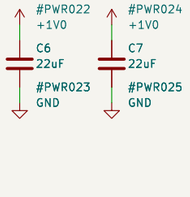


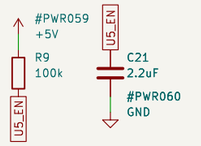


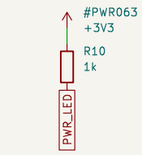






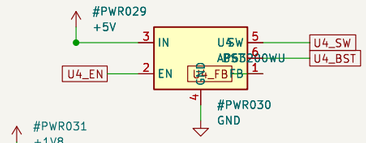


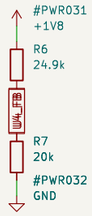


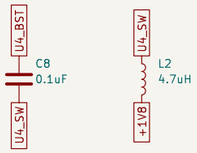


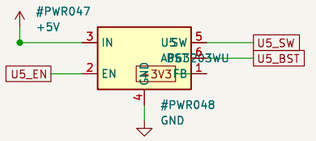


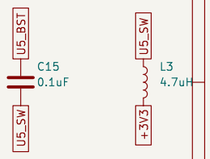






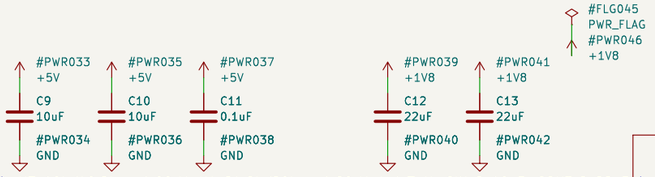


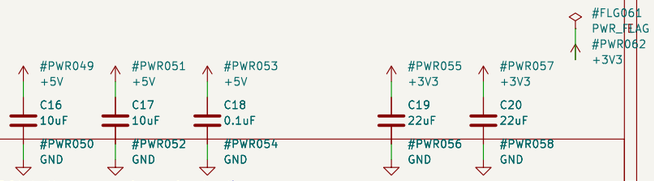






H1-H4 match the carrier board holes (M3 standoffs, 11 mm between the boards). H5-H7: main board only.





Vout = 0.8 V x (1 + Rtop/Rbot): U3 10k/40.2k = 1.00 V, U4 24.9k/20k = 1.80 V. L 4.7uH, CIN 2x10uF, COUT 2x22uF (+ FPGA bulk caps on fpga_cfg sheet).

Quad HDMI test pattern generator — FPGA main board (XC7A35T, carrier stacks on top)

gyuhooo

Sheet: /Power: USB-C 5 V in, 1.0 V / 1.8 V / 3.3 V bucks/

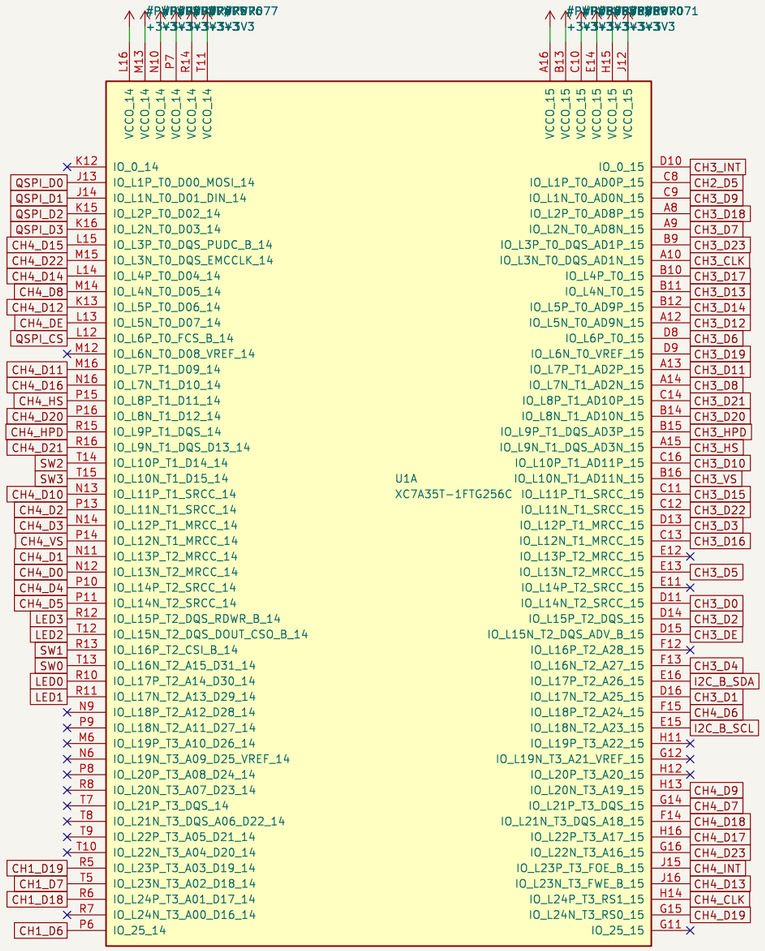
File: power.kicad_sch

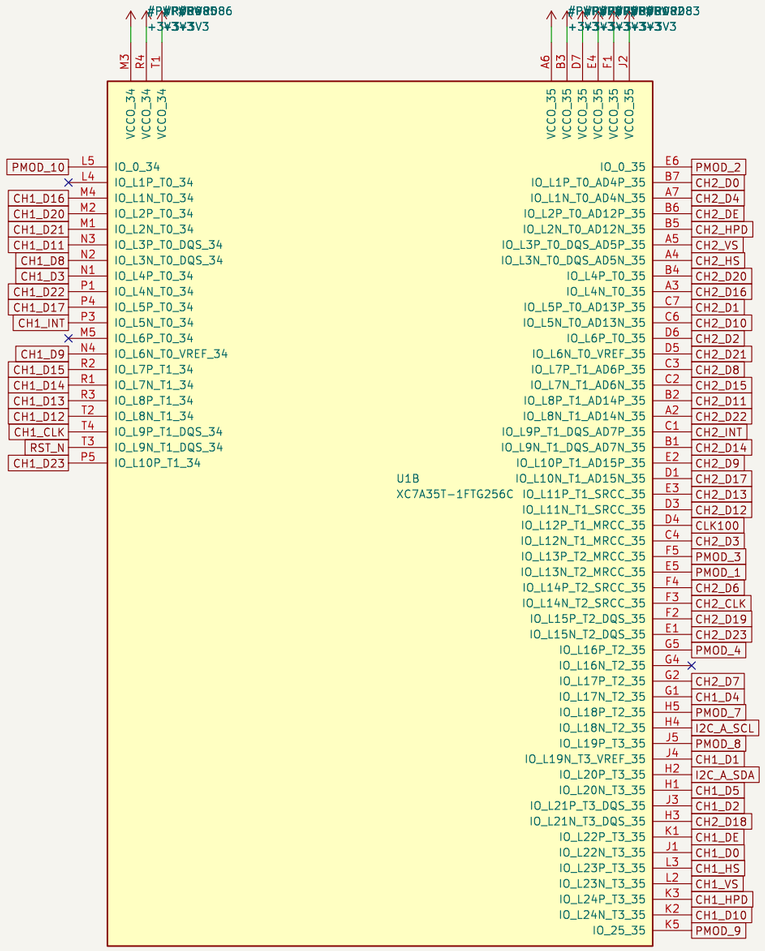
Title: Power: USB-C 5 V in, 1.0 V / 1.8 V / 3.3 V bucks

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XC7A35T-1FTG256C I/O banks. All banks VCCO = 3.3 V (LVCMOS33). CHn_* nets go to the carrier sockets J1/J2 (see fpga_pinmap.csv).

Assignment is generated by hardware/gen/fpga_common.py: BGA escape plan (ring 1–2 on top, ring 3–4 on In2, deeper on In3) and monotonic perimeter-order mapping of the 124 header signals so that the fan-out to the sockets does not cross. NC = unused I/O (left open).





Quad HDMI test pattern generator — FPGA main board (XC7A35T, carrier stacks on top)

gyuhooo

Sheet: /FPGA I/O banks 14/15/34/35/
File: fpga_io.kicad_sch

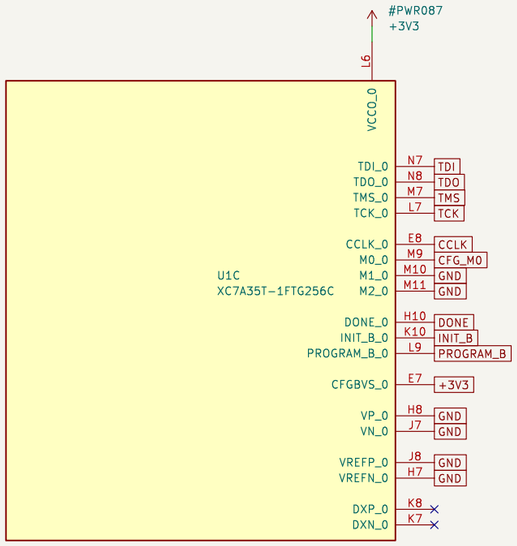
Title: FPGA I/O banks 14/15/34/35

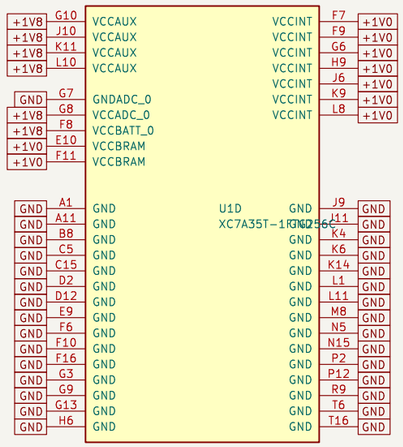
Size: A3	Date: 2026-09-14	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/6

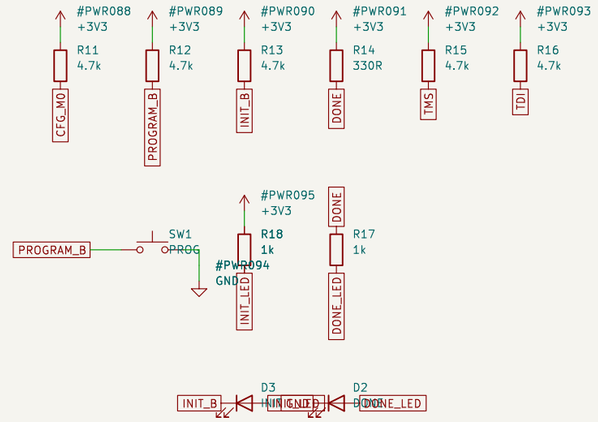
Configuration: Master SPI x4 from W25Q32JV (M[2:0] = 001: M0 pulled up, M1/M2 to GND). CFGBVS = VCCO_0 = 3.3 V. JTAG: 2x7 2 mm (J3) and 1x6 (J4) in parallel.

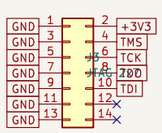
DONE: 330R pull-up + green LED. INIT_B: 4.7k pull-up + red LED (lit while low = not configured / CRC error). PROGRAM_B: 4.7k pull-up + push button SW1.

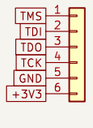
XADC unused: VREFP/VREFN/VP/VN/GNDADC to GND, VCCADC and VCCBATT to +1V8, DXP/DXN open. VCCBRAM tied to VCCINT (1.0 V, allowed for -1 speed grade).

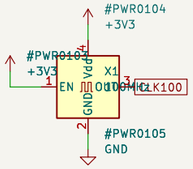


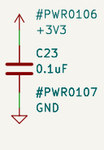


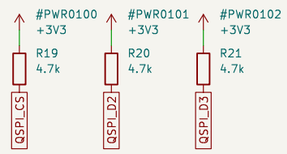


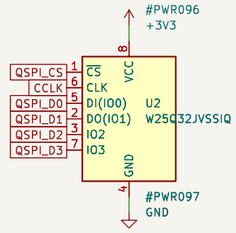


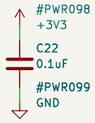






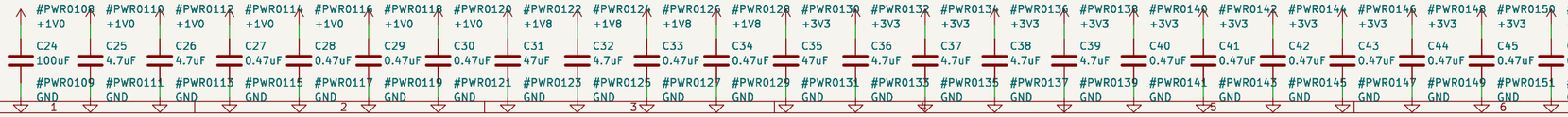






CLK100 -> D4 (IO_L12P_T1_MRCC_35). hdl/top.v clk100; MMCM per the W25Q32JV (3.3 V) -> 17.5 MHz (set_bitstream_config -bpi_sync_mode). CONFIG.SPI_BUSWIDTH 4, CONFIGRATE 33. CCLK/D0-D3/CS are the bank-14 dual-purpose p

FPGA decoupling (bottom side under/around the BGA): +1V0 100u + 2x4.7u + 4x0.47u, +1V8 47u + 4.7u + 2x0.47u, +3V3 (VCCO) 47u + 4x4.7u + 9x0.47u.



Quad HDMI test pattern generator — FPGA main board (XC7A35T, carrier stacks on top)

gyuhooo

Sheet: /FPGA configuration, power pins, QSPI flash, JTAG, clock/

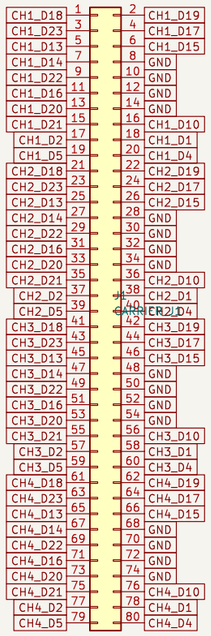
File: fpga_cfg.kicad_sch

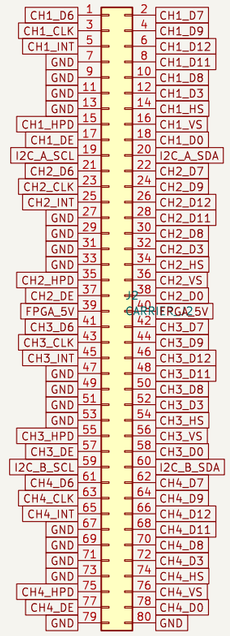
Title: FPGA configuration, power pins, QSPI flash, JTAG, clock

Size: A3	Date: 2026-09-14
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	Id: 4/6

J1/J2: 2x40 female sockets (2.54 mm, THT) at the mirror positions of the carrier's bottom-side headers J7/J8. Pin n of J1 = carrier J7 pin n, J2 = J8.

124 signals (3.3 V LVCMOS, up to 148.5 MHz SDR), 34 GND, FPGA_5V on J2 pins 39/40 (optional 5 V link between the boards, JP1 here and JP1 on the carrier).







JP1 open (default): each board has its own USB-C supply. Close JP1 on both boards to power the pair from one USB-C (max 2 A total).

Quad HDMI test pattern generator — FPGA main board (XC7A35T, carrier stacks on top)

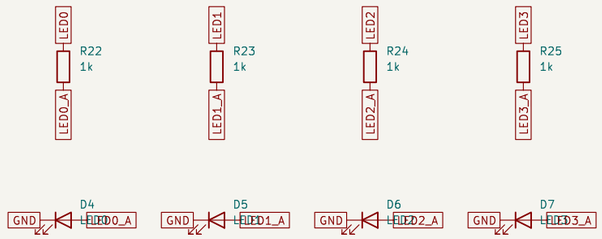
gyuhooo

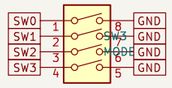
Sheet: /Carrier board sockets J1/J2 (mate quad_hdmi_tx J7/J8)/
File: carrier_if.kicad_sch

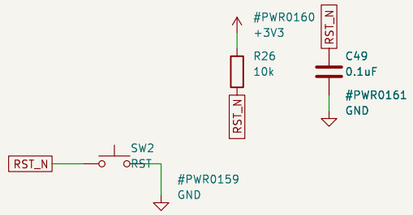
Title: Carrier board sockets J1/J2 (mate quad_hdmi_tx J7/J8)

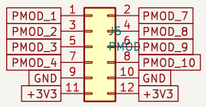
Size: A3	Date: 2026-09-14	**Rev: A**
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LED0–3 (hdl/top.v led[3:0]: channel init done), SW0–3 DIP (spare, internal pull-ups), SW2 reset (RST_N, 10k pull-up), PMOD J5 (8 spare I/Os, bank 35).









SW0–3: closed = low. Enable PULLUP in the XDC. Not used by the current HDL (reserved for the 2-input mode select).

PMOD: header pin 1,3,5,7 = Pmod 1–4; 2,4,6,8 = Pmod 7–10; 9/10 = GND; 11/12 = VCC (3.3 V, 100 mA max).

Quad HDMI test pattern generator — FPGA main board (XC7A35T, carrier stacks on top)

gyuhooo

Sheet: /User I/O: LEDs, DIP switch, reset button, PMOD/

File: io_misc.kicad_sch

Title: User I/O: LEDs, DIP switch, reset button, PMOD

Size: A3	Date: 2026-09-14	**Rev: A**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/6